

//SIT 55 : ASSIGNMENT 8 : FILE HANDLING

/*Implement a program for maintaining a database of student records using Files.

Student has Student_id,name, Roll_no, Class, marks and address. Display the data for few students.

1. Create Database , 2. Display Database , 3. Delete Records , 4. Update Record , 5. Search Record*/

```
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.io.InputStreamReader;
import java.io.PrintWriter;

class StudentRecords
{
    static BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

    public void addRecords() throws IOException
    {
        // Create or Modify a file for Database
        PrintWriter pw = new PrintWriter(new BufferedWriter(new FileWriter("studentRecords.txt",true)));
        String name, Class, fname, mname, address, dob;
        int age;
        long telephoneNo;
        String s;
        boolean addMore = false;
        // Read Data
        do
        {
            System.out.print("\nEnter NAME: ");
            name = br.readLine();

            System.out.print("Enter FATHER NAME: ");
            fname = br.readLine();

            System.out.print("Enter MOTHER NAME: ");
            mname = br.readLine();

            System.out.print("Enter ADDRESS: ");
            address = br.readLine();

            System.out.print("Enter CLASS: ");
            Class = br.readLine();

            System.out.print("Enter DATE OF BIRTH (dd/mm/yy) : ");
            dob = br.readLine();

            System.out.print("Enter AGE: ");
            age = Integer.parseInt(br.readLine());

            System.out.print("Enter TELEPHONE NO.: ");
            telephoneNo = Long.parseLong(br.readLine());
            // Print to File
```

```
pw.println(name);
pw.println(fname);
pw.println(mname);
pw.println(address);
pw.println(Class);
pw.println(dob);
pw.println(age);
pw.println(telephoneNo);
```

```
System.out.print("\nRECORDS ADDED SUCCESSFULLY ! \n\n DO YOU WANT TO ADD MORE RECO
RDS (Y/N) : ");
s = br.readLine();
if(s.equalsIgnoreCase("Y"))
{
    addMore = true;
    System.out.println();
}
else
    addMore = false;
}
while(addMore);
pw.close();
showMenu();
}
public void readRecords() throws IOException
{
    try
    {
        // Open the file
        BufferedReader file = new BufferedReader(new
        FileReader("studentRecords.txt"));
        String name;
        int i=1;
        // Read records from the file
        while((name = file.readLine()) != null)
        {
            System.out.println("S.No. : " +(i++));
            System.out.println("-----");
            System.out.println("\nNAME: " +name);
            System.out.println("FATHERS'S NAME : "+file.readLine());
            System.out.println("MOTHER'S NAME : "+file.readLine());
            System.out.println("ADDRESS: "+file.readLine());
            System.out.println("CLASS: "+file.readLine());
            System.out.println("DATE OF BIRTH : "+file.readLine());
            System.out.println("AGE : "+Integer.parseInt(file.readLine()));
            System.out.println("TEL. NO.: "+Long.parseLong(file.readLine()));
            System.out.println();
        }
        file.close();
        showMenu();
    }
    catch(FileNotFoundException e)
    {
        System.out.println("\nERROR : FILE NOT FOUND !!!");
    }
}
```

```

}
public void clear() throws IOException
{
// Create a blank file
PrintWriter pw = new PrintWriter(new BufferedWriter(new
FileWriter("studentRecords.txt")));
pw.close();
System.out.println("\n ALL RECORDS CREATED SUCCESSFULLY !");
for(int i=0;i<999999999;i++); // Wait for some time
showMenu();
}
public void showMenu() throws IOException
{
System.out.print("\n 1 : ADD RECORDS \n 2 : DISPLAY RECORDS \n 3 : CLEAR ALL RECORDS \n 4 :
EXIT \n\n YOUR CHOICE : ");
int choice = Integer.parseInt(br.readLine());
switch(choice)
{
case 1:
addRecords();
break;
case 2:
readRecords();
break;
case 3:
clear();
break;
case 4:
System.exit(1);
break;
default:
System.out.println("\n INVALID CHOICE !");
break;
}
}
}
public static void main(String args[]) throws IOException
{
StudentRecords call = new StudentRecords();
call.showMenu();
}
}

```

/*OUTPUT:

```

1 : ADD RECORDS
2 : DISPLAY RECORDS
3 : CLEAR ALL RECORDS
4 : EXIT

```

YOUR CHOICE : 1

```

ENTER NAME: ABC
ENTER FATHER NAME: AA
ENTER MOTHER NAME: BB

```

ENTER ADDRESS: CC
ENTER CLASS: FE
ENTER DATE OF BIRTH (dd/mm/yy) : 22/02/2005
ENTER AGE: 19
ENTER TELEPHONE NO.: 123

RECORDS ADDED SUCCESSFULLY !

DO YOU WANT TO ADD MORE RECORDS (Y/N) : Y

ENTER NAME: PQR
ENTER FATHER NAME: PP
ENTER MOTHER NAME: QQ
ENTER ADDRESS: RR
ENTER CLASS: SE
ENTER DATE OF BIRTH (dd/mm/yy) : 18/01/2004
ENTER AGE: 20
ENTER TELEPHONE NO.: 456

RECORDS ADDED SUCCESSFULLY !

DO YOU WANT TO ADD MORE RECORDS (Y/N) : N

1 : ADD RECORDS
2 : DISPLAY RECORDS
3 : CLEAR ALL RECORDS
4 : EXIT

YOUR CHOICE : 2
S.No. : 1

NAME: ABC
FATHERS'S NAME : AA
MOTHER'S NAME : BB
ADDRESS: CC
CLASS: FE
DATE OF BIRTH : 22/02/2005
AGE : 19
TEL. NO.: 123

S.No. : 2

NAME: PQR
FATHERS'S NAME : PP
MOTHER'S NAME : QQ
ADDRESS: RR
CLASS: SE
DATE OF BIRTH : 18/01/2004
AGE : 20
TEL. NO.: 456

- 1 : ADD RECORDS
- 2 : DISPLAY RECORDS
- 3 : CLEAR ALL RECORDS
- 4 : EXIT

YOUR CHOICE :*/